Case Studies: Sports Analytics

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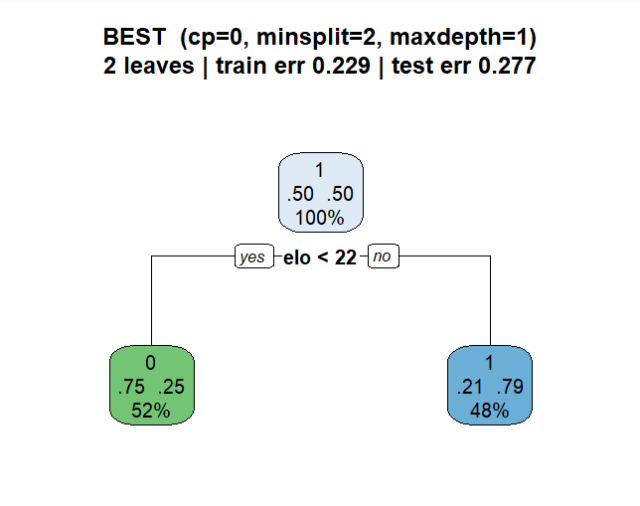
October 29, 2026

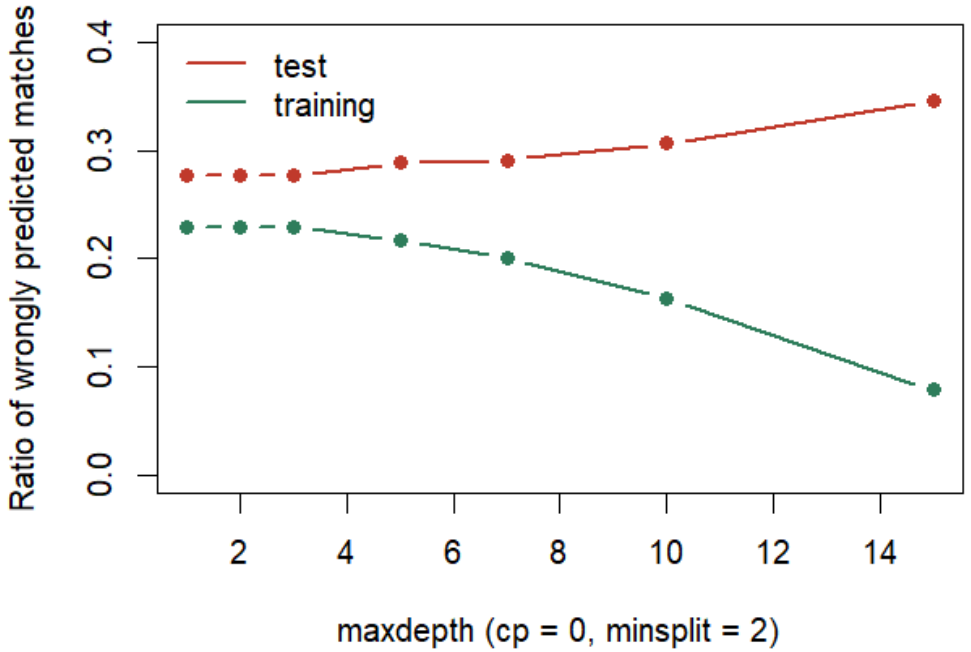
Tennis Classification Tree

Highlight of Tennis Grid Search Model Comparison

comment	cp	minsplit	maxdepth	train_error	test_error	nsplits
least test error	0.000	2	1	0.229	0.277	1
reasonable performer using more than one split	0.001	40	5	0.224	0.279	6
worst overfit	0.000	2	15	0.079	0.345	434

Winner is a **single split on elo**: 72.3% of unseen matches correct (logistic regression 73.1%, coin flip 50%).





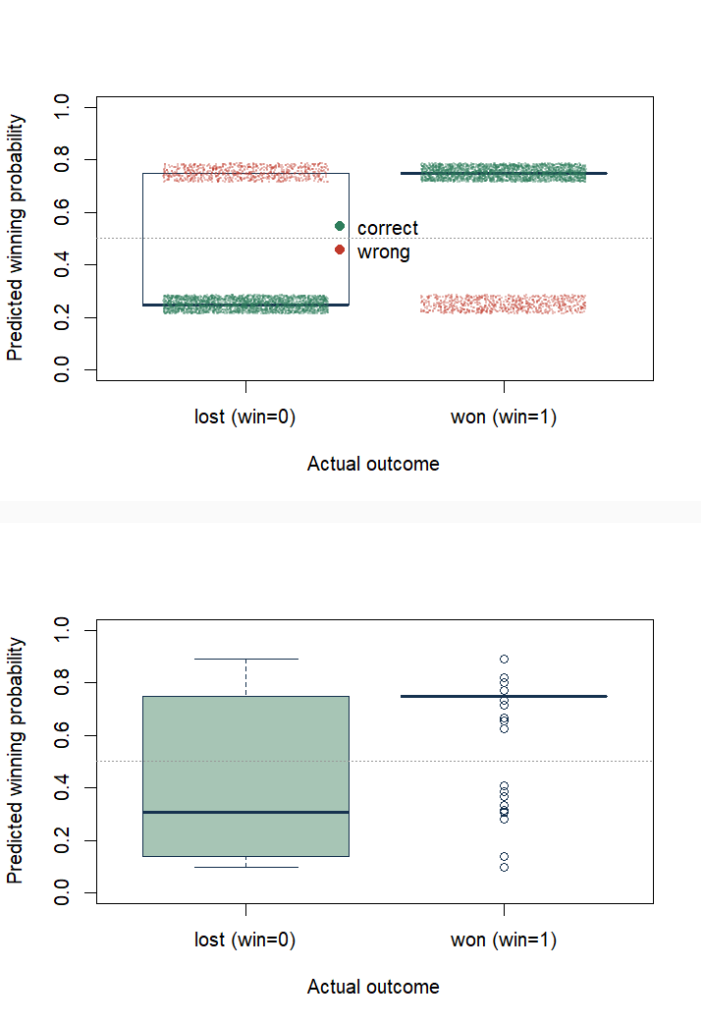
Tennis: Why the Tree Stays So Small

Default (cp = 0.01)

- 1 split, uses `elo` only
- only **2 probabilities**: 0.249 / 0.750

cp = 0.001

- 22 leaves, **19 probabilities**
- uses **all four** predictors



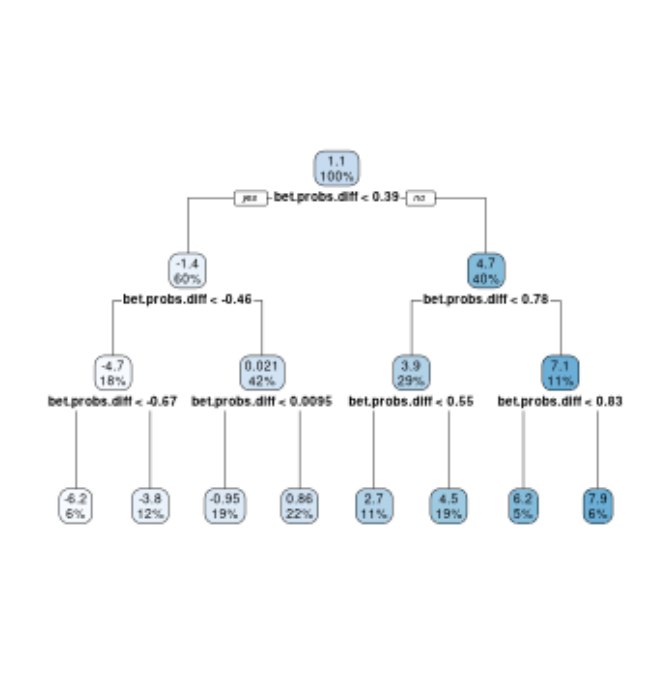
Predicted winning probabilities by outcome: 2 values (top, cp = 0.01) vs. 19 values (bottom, cp = 0.001).

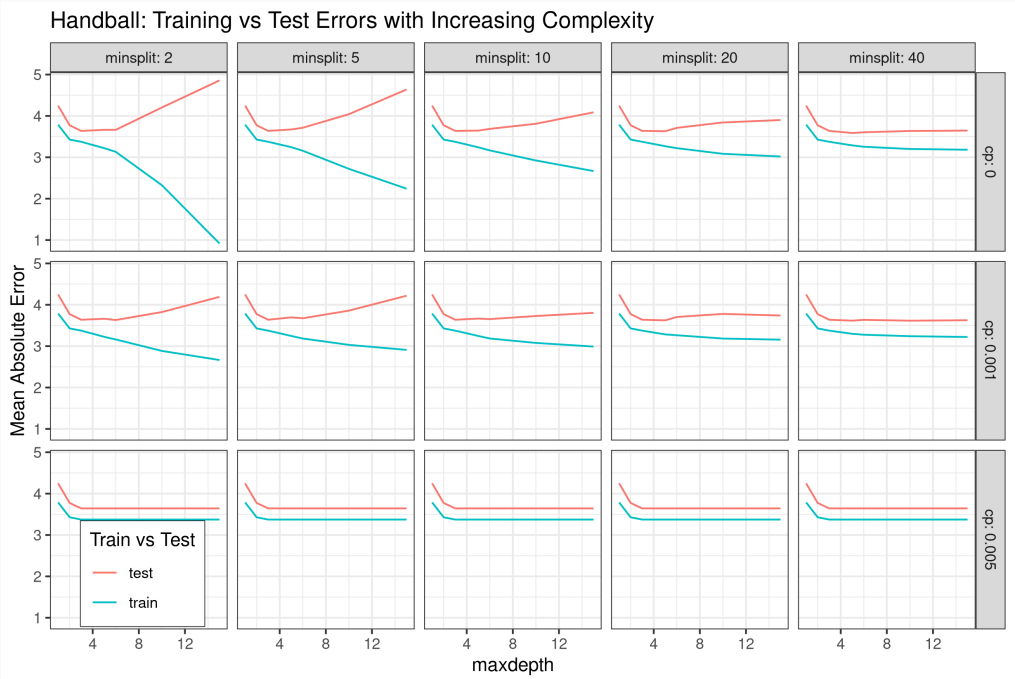
Handball Tree Regression

Highlight of Handball Grid Search Model Comparison

comment	cp	minsplit	maxdepth	train_mae	train_rmse	test_mae	test_rmse	nsplits
best test mae	0	40	5	3.288	4.213	3.588	4.582	25
compromise btw number of splits & test mae	0	2	3	3.375	4.307	3.638	4.606	7
worst overfit in mae & rmse	0	2	15	0.916	1.915	4.859	6.387	914

Best compromise:



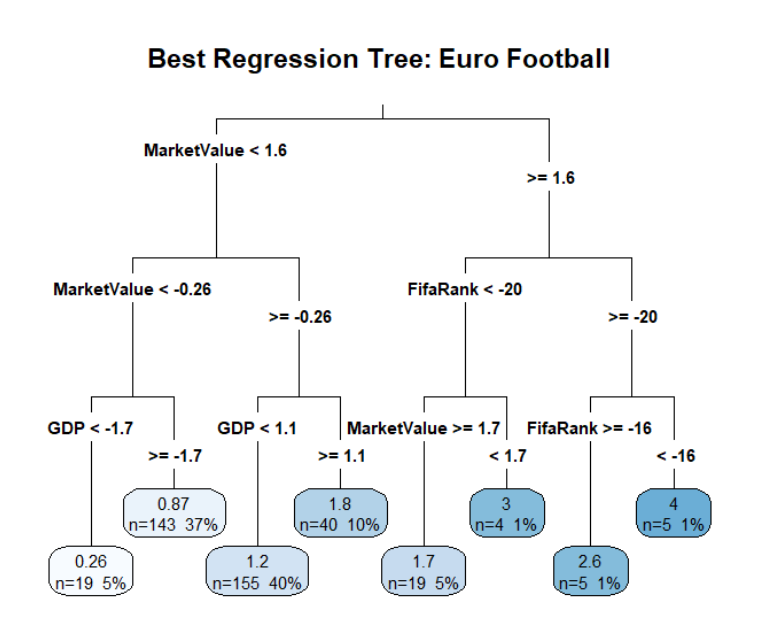


Football Regression Trees: Model Selection

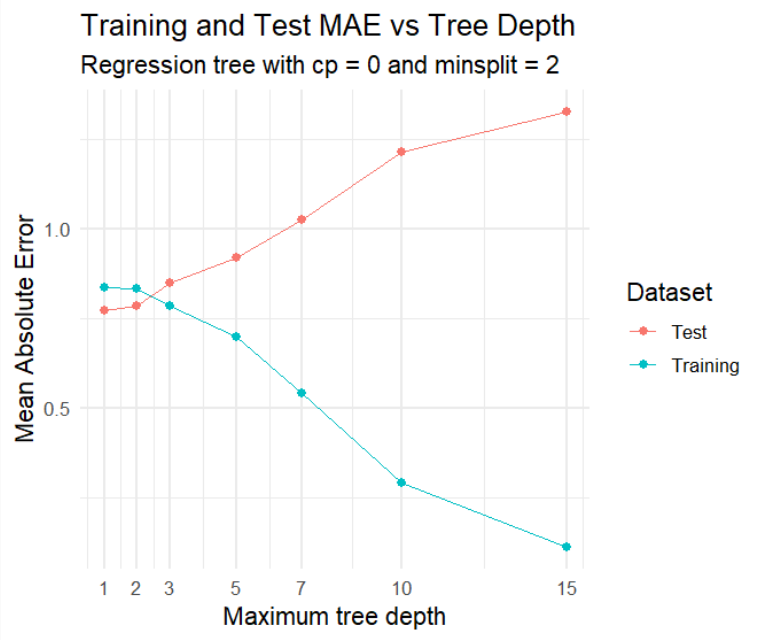
Tree specifications and predictive performance

Tree	cp	minsplit	maxdepth	Train MAE	Test MAE	Test RMSE
Baseline tree	0.01	20	30	0.779	0.784	1.055
Best tree	0	5	3	0.780	0.741	0.985
Worst overfit tree	0	2	15	0.112	1.326	1.716

Best regression tree

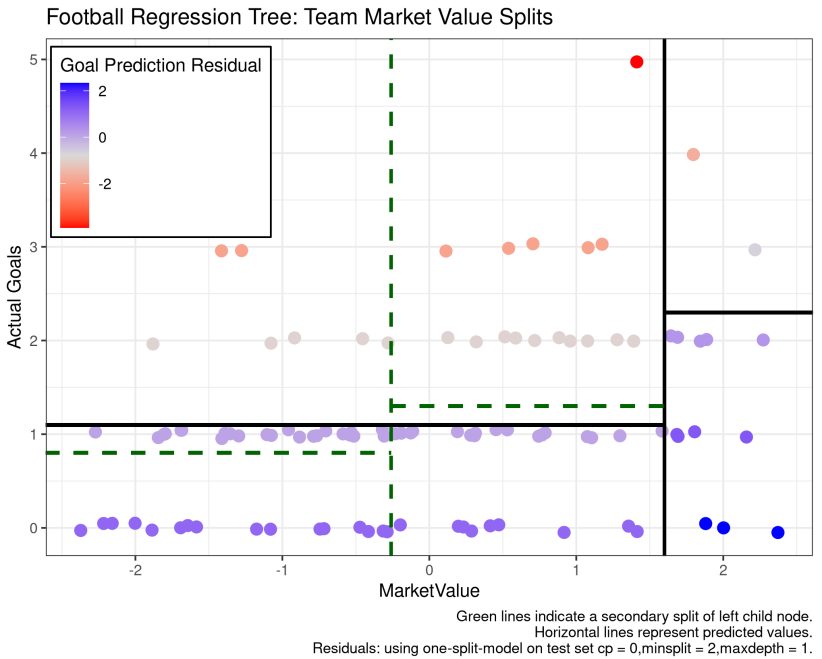


Training vs. test error

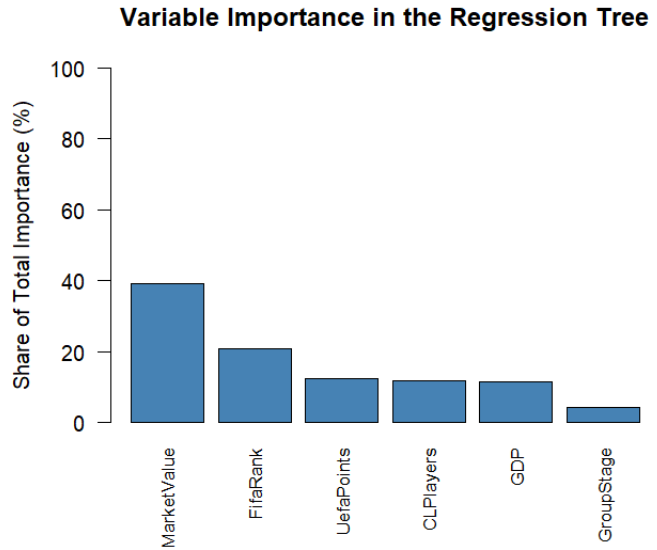


Football Regression Trees: Interpretation

Splits based on market value



Variable importance



Predictive performance on Euro 2024 test data

Model	Test MAE	Test RMSE
Baseline model	0.781	1.006
Best regression tree	0.741	0.985
Lasso regression	0.735	0.946